

Chongkun Deng

cd3411@columbia.edu | ORCID: 0009-0005-0808-9278

Research Interests

Ordinal and comparison-based methods; metric geometry and tree metrics; high-dimensional probability and statistics; Euclidean and hyperbolic geometry; algorithms and optimization for geometric inference.

Education

Columbia University, Graduate School of Arts and Sciences

September 2023 - December 2024

M.A. in Statistics; GPA: 3.82/4.00

University of California San Diego

September 2019 - June 2023

B.S. in Mathematics; GPA: 3.69/4.00; Major GPA: 3.67/4.00

Preprints

1. Farthest-cell triplet entropy: high-dimensional shell limits and hyperbolic curvature amplification.

arXiv:2609.02362, 2026.

Introduces a comparison-based entropy statistic for the farthest-prototype label among random triples. Derives high-dimensional limits for isotropic Euclidean and hyperbolic radial models and studies model-calibrated curvature inference. It also clarifies the need for an external length unit to identify absolute curvature and the sensitivity to angular anisotropy.

2. Tree Recognition and Graph Edge Minimization from Anchored Distance Comparisons.

Preprint, September 2026.

Studies when anchored comparisons $d(o, x) < d(o, y)$ admit shortest-path realizations. Gives two quadratic-time constructions that recover any positive tree realization, followed by a sparse linear program for the weights. For general graphs, proves non-joint edge-omission phenomena, a linear gap result, and the sharp $n(n-1)/2 - 1$ extremal bound; minimum-edge complexity remains open.

Research Experience

Independent Research

August 2023 - present

Comparison-based geometry, tree recognition, and graph realization

- Develop methods for recovering metric, geometric, and combinatorial structure when numerical distances are unavailable.
- Combine probabilistic limit theory, metric geometry, graph algorithms, linear programming, and exact computational verification.

RTG Reading Group

March 2022 - June 2022

University of California San Diego

- Studied advanced mathematical theory under the supervision of graduate students.
- Investigated the correspondence between algebraic curves and discrete valuation rings and presented theoretical material in a collaborative reading group.

Teaching Experience

Teaching Assistant, University of California San Diego

September 2020 - March 2021

MATH 20C: Calculus and Analytic Geometry for Science and Engineering

- Assisted with the course for two quarters; led discussion sections and office hours and graded assignments and examinations.
- Prepared course materials and contributed to discussions about course design.

Honors and Competitions

Harvard-MIT Mathematics Tournament, General Round

November 2018

Top 3%, rank 24.

Technical Skills and Languages

Programming: Python, R

Languages: Chinese (native), English (fluent), Japanese (conversational)